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ORIGINAL PAPER

Determination of Poly- and Perfluorinated Alkyl Substances (PFAS) in Livestock and Game Animals in Ghana: First Data on Contamination in the Liver Tissue of Mammals and Birds from Africa

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Abstract

Per- and polyfluoroalkyl substances (PFAS) are anthropogenic pollutants that have been shown to accumulate in the livers of animals. For this reason, liver tissue from livestock and game can be used as an indicator of PFAS contamination in the environment and the potential risk of human exposure. In this study, we present data on the contamination of livestock and game animals from Ghana for the first time. Liver samples from 14 different species were investigated for 21 PFAS. In total 11 different PFAS could be detected in 85% of the samples, with perfluorohexanoic acid showing the highest concentrations (22 µg/kg) and being the most frequently detected (75%) substance. The median value for the sum of the quantified PFAS was 1.25 µg/kg. Significantly more PFAS were detected in the liver samples from omnivorous than in those from herbivorous animals, and although the differences in the overall concentration between those groups were not significant, omnivorous animals showed significantly higher concentrations for long-chain perfluorocarboxylic acids. No significant differences were found between livestock and game animals. An assessment of PFAS levels in muscle meat was conducted to determine whether there is evidence of increased exposure of the population through the consumption of animal-based food.

Keywords PFOS · PFOA · Persistent organic pollutant · Environmental contamination · Food safety

Introduction

Poly- and perfluorinated alkyl substances (PFAS) are a class of anthropogenic organic substances that have emerged as some of the most concerning chemical pollutants of the twenty-first century. The replacement of some or all carbon-bound hydrogen atoms with fluorine gives PFAS valuable technical properties like oil- and water-repellency, which lead to a widespread use both in industrial applications and in consumer products. Their unique chemical properties also make them resistant to chemical and microbial degradation, thereby enabling them to function as persistent organic pollutants (Glüge et al. 2020; Gaines 2023). PFAS have shown to accumulate in the environment and the food chain, making them detectable even in animals in remote areas such as the Arctic Circle (Boisvert et al. 2019). The European Food Safety Authority (EFSA) has conducted a re-evaluation of PFAS and their tolerable weekly intake through food and recommends limiting the intake of perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorohexanesulfonic acid (PFHxS) and perfluorooctanesulfonic acid (PFOS) to 4.4 ng/kg body weight per week (EFSA Panel on Contaminants in the Food Chain et al. 2020).

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Many studies have been conducted to investigate the uptake of PFAS by livestock. For cattle, both the uptake of PFAS into plasma and skin from contaminated feed and water, as well as the transfer of PFOS into dairy milk, have been reported (van Asselt et al. 2013; Lupton et al. 2022). For pigs and chickens, the uptake and transfer into the tissue as well as into eggs has also been reported before (Mikolajczyk et al. 2022, 2024; Draghi et al. 2025).

Among wild living animals, the European wild boar (*Sus scrofa*) is among the best studied species. Their omnivorous diet and their habit of digging through the top layer of the soil in search for food leads to a usually high burden of PFAS in the liver of the animals (Stahl et al. 2012; Barola et al. 2020; Rupp et al. 2023). They have therefore been proclaimed as bioindicators for environmental PFAS contamination (Kowalczyk et al. 2018).

PFAS contamination on the African continent has been studied far less extensively than in Europe. The available literature focuses on marine animals (Chokwe et al. 2024; Umejuru et al. 2024). In Ghana, there is a study on the contamination of fish in the Gulf of Guinea. In their study, the authors found no significant difference between the study sites in Ghana and only noted in general that the contamination of fish was similar to that found in other studies from Brazil, Tunisia, and China (Ekperusi et al. 2023). Studies on blood serum and breast milk contamination have been conducted to estimate the exposure of the population in Ghana. The authors of the breast milk study describe the results as consistent with other international publications and identified PFOS and PFOA as the two dominant substances (Jonathan et al. 2024). Furthermore, PFOS and PFOA were identified as the predominant substances in blood serum, along with PFHxS (Dartey et al. 2021).

To date, little data on PFAS in African wildlife has been published. To our knowledge, no data on contamination in birds and mammals has been reported. The aim of our study is to help close this gap by presenting a first overview of PFAS contamination in the livers of mammals and birds from Ghana. This will give insight into environmental exposure and the possible implications for local human exposure through animal-based foods.

Materials and Methods

Study Area

This study was conducted in the Kumasi Metropolis, located in the Ashanti Region of Ghana. It is situated between latitude 6.35° N and 6.40° N, and longitude 1.30° W and 1.37° W, covering an area of approximately 299 km^2. It is a major commercial, industrial, and cultural hub, located approximately 500 km north of the Equator and 200 km north of the Gulf of Guinea (Darko et al. 2017). Within the metropolis, the Kumasi Main Abattoir and the Atwemonom Game Market were selected for the sampling of liver tissues of domestic animals and wild animals, respectively. These sites are known to be the central point or hub for purchasing, slaughtering, and distributing meat to satellite markets within and around the metropolis. In addition, Abrepo Junction, which is a busy, vibrant suburb of the metropolis, was selected for sampling the livers of chickens and pigs at a local butcher. Figure 1 shows the locations of the sampling sites within the city of Kumasi. Detailed descriptions of the Kumasi Main Abattoir and the Atwemonom Game Market can be found in the supplementary information.

Sample Collection

Liver samples from different animals were collected from the study sites. In total, 41 liver samples from 14 different animal species were obtained from October to December 2023. Information on the species sampled can be found in Table 1, detailed information regarding the individual samples is given in Table S4. Either approximately 1 kg of liver was purchased per animal (cattle, goats, and sheep), or the entire liver of the animal was purchased if it weighed less than 1 kg due to the animal's low body weight. All samples were carefully placed and sealed in their respective prelabelled PFAS-free PE-bags. All samples were purchased fresh in the morning after the animals were slaughtered, placed on ice, and transported to the laboratory the same day. The samples were then stored at—20 °C until further preparation.

Chemicals and Materials

A PFAS mixed standard (EU-5813-NSS from Wellington Laboratories Inc., Guelph, Canada) was used for quantification and quality control samples. This contained perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA) perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), PFOA, PFNA, perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), perfluorododecanoic acid (PFDoDA), perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA), perfluorobutanesulfonic acid (PFBS), perfluoropentansulfonic acid (PFPeS), PFHxS, perfluoroheptanesulfonic acid (PFHpS), PFOS, perfluorononansulfonic acid (PFNS), perfluorodecanesulfonic acid (PFDS), perfluoroundecansulfonic acid (PFUnDS), perfluorododecansulfonic acid (PFDoDS) and perfluorotridecansulfonic acid (PFTrDS). An isotopically labelled mix standard (MPFAC-C-ES from Wellington Laboratories Inc., Guelph, Canada) consisting of ^{13}C-labelled PFAS was used



Fig. 1 Locations of the Kumasi Main Abattoir (1), the Atwemonom Game Market (2) and the Abrepo Junction (3) within the city of Kumasi, Ghana. Map source: © OpenStreetMap contributors and available from https://www.openstreetmap.org

as an internal standard for all samples and calibrators (see Table S3). Water, methanol and acetonitrile were purchased from Carl Roth GmbH + Co. KG (Karlsruhe, Germany) and acetone, NH_3-solution (25%), formic acid and ammonium acetate were purchased from VWR International GmbH (Darmstadt, Germany). All chemicals met grade requirements for high performance liquid chromatography-mass spectrometry.

Sample Preparation

Homogenization and Extraction

The collected liver samples were homogenized at the Kumasi Lab, using a blender. The livers were placed in the blender without any additional liquid and homogenized until they were blended into a smooth paste. The blender was thoroughly cleaned with deionized water followed by methanol between each sample to prevent contamination. Afterwards, 500 mg of the sample were weighed into a polypropylene

Table 1 Overview of the species examined and the number of liver samples taken (n) from game and livestock from Kumasi, Ghana

Name	Species	Point of purchase	n
Ahanta francolin	*Pternistis ahantensis*	Atwumunom game market	3
Bay Duiker	*Cephalophus dorsalis*	Atwumunom game market	3
Black Duiker	*Cephalophus niger*	Atwumunom game market	3
Cattle	*Bos taurus*	Kumasi Abattoir	3
Chicken	*Gallus domesticus*	Abrepo Junction	3
Civet	*Civettictis civetta*	Atwumunom game market	3
Goat	*Capra hircus*	Kumasi Abattoir	2
Grasscutter	*Thryonomys swinderianus*	Atwumunom game market	3
Ground Squirrel	*Euxerus erythropus*	Kumasi Abattoir	3
Maxwell`s Duiker	*Philantomba maxwellii*	Atwumunom game market	3
Pig	*Sus scrofa domesticus*	Abrepo Junction	3
Royal antelope	*Neotragus pygmaeus*	Atwumunom game market	3
Sheep	*Vvis aries*	Kumasi Abattoir	3
Wild hog	*Potamochoerus porcus*	Atwumunom game market	3

tube to which 10 mL methanol was added and left overnight at room temperature. The next day, 1.5 mL of the supernatant were transferred to a polypropylene vial and shipped to the lab in Germany for cleanup and analysis.

Solid Phase Extraction for the Quantification of PFAS

The cleanup consisted of a solid phase extraction, based on the method published prior to this study (Felder et al. 2023). The extraction was performed using Chromabond HR-XAW cartridges (Macherey & Nagel, Düren, Germany) according to DIN 38414—14. The cartridges were each conditioned with 2 mL methanol + 0.1% formic acid, methanol, and water. Subsequently, 1 mL of the sample extract was mixed with 1 mL water + 0.1% formic acid and 50 µL internal standard (0.02 ppm) in the cartridge. After sample addition, the cartridges were rinsed with 2 mL of water + 0.1% formic acid, acetone:acetonitrile (1:1) + 1% formic acid, and methanol. The samples were eluted using 3 mL methanol containing 0.1% NH_3. The eluate was filtered through a Chromabond Carbon A cartridge (Macherey & Nagel, Düren, Germany), evaporated to dryness, and reconstituted with 1 mL methanol:water (4:6) in preparation for analysis via high performance liquid chromatography-tandem mass spectrometry (HPLC–MS/MS).

Sample Preparation for the Confirmation of PFBA and PFPeA Results

Since PFBA and PFPeA only show one characteristic mass transition during fragmentation in the MS/MS measurement performed here, the quantification results must be confirmed by a second measurement using a different chromatographic separation. For this purpose, a second aliquot was prepared independently. For this, 100 µL of the sample extract were diluted with 100 µL of an aqueous internal standard solution (100 ppb). The samples were then used for measurement without further preparation.

Instrumental Analysis

Chromatography

For chromatographic separation, an Agilent 1290 Infinity II system (Agilent Technologies GmbH, Waldbronn, Germany) was used. The separation for the quantification of the complete analyte panel was carried out with a ZORBAX Eclipse Plus C18 2.1 × 100 mm 1.8 Micron column (Agilent Technologies GmbH, Waldbronn, Germany) and a ZORBAX Eclipse Plus C18 2.1 × 5 mm 1.8 Micron Guard column, tempered to 35 °C in a column oven. The separation for the confirmation of PFBA and PFPeA was performed on a NUCLEODUR PFAS 3 µm 2 × 100 mm column (Macherey & Nagel-Düren, Germany) combined with a NUCLEOSHELL RP18 Plus 2.7 µm 2 × 4 mm Guard column (Macherey & Nagel-Düren, Germany), also tempered to 35 °C in a column oven. Further information on chromatographic separations can be found in Table S1 and Table S2.

Mass Spectrometry

Mass spectrometric measurements were performed on a Sciex Qtrap 6500 + System (AB Sciex, Darmstadt, Germany) with electrospray ionization performed in negative mode. The injection volume for each method, both for quantifying the whole PFAS spectrum and for confirming the results for PFBA and PFPeA, was 100 µL. Detection was performed in dynamic multi-reaction-monitoring mode, with two characteristic mass transitions for each analyte. The exceptions were PFBA and PFPeA, for which only one characteristic transition could be used, which is due to the short length of their carbon chains. The mass transitions for all analytes and internal standards are listed in Table S3. The Sciex OS Analytic module (Sciex AB, Darmstadt, Germany) was used for instrument operation and quantitative evaluation. During the evaluation of the results, all peaks were checked manually. The applied criteria included the

quality of peak integration, peak retention times compared to the calibration curve and isotope-labelled internal standards, and the ion ratio of the two mass transitions of the substances. Only peaks with a maximum deviation of 1% from the calibration curve in terms of retention time ratio of analyte and corresponding internal standard and an ion ratio of 30% compared to the standards were included in the evaluation. This procedure follows the recommendations of the European Union Reference Laboratory for halogenated POPs in Feed and Food (EURL POPs) (EURL for halogenated POPs in Feed and Food 2024) The limit of quantification (LOQ) for both methods was 0.2 µg/kg for each substance and the limit of detection (LOD) was 0.07 µg/kg. The limit of quantification was determined in accordance with the EURL POPs guidelines, using the lowest validated level approach (EURL for halogenated POPs in Feed and Food 2024). The limit of detection was determined by a signal to noise ratio of 3:1. For quantification an isotopically labelled PFAS was assigned to each substance, recovery rates ranged between 84% (PFDS) and 109% (PFBA). The performance of the quantification method was tested as part of a European proficiency study for halogenated organic pollutants in pork liver, organized by the EURL POPs. The method met the performance criteria with all z-scores being below 1.

Quantification and Quality Control

Seven calibrators, covering a concentration range from 4 ng/L to 2000 ng/L, were used to quantify the results. All calibrators contained the internal standard at a concentration of 1000 ng/L. For all analytes, calibration curves were generated based on the area ratios of the peaks of the analyte and the corresponding internal standard. For samples in which individual calculated concentrations lay above the calibration curve, additional dilutions were prepared and measured to shift the concentration in the measured sample extract into the range of the calibration curve. Extraction recovery samples and blank samples were prepared alongside the samples to ensure proper sample preparation and to prevent undetected contamination within the extracts.

Statistical Methods

IBM SPSS Statistics 29.0 (IBM Corp., Armonk, USA) was used for the statistical analysis of the results. The lower bound approach was chosen for the analysis. Therefore, all concentrations that were < LOQ but > LOD were assigned a value of 0.07 µg/kg (LOD). All values < LOD were considered zero. The Mann–Whitney U test was used to test for statistical differences, with the significance level of $\alpha = 0.05$.

Results

PFAS Concentrations

Overall, out of the 21 substances measured, 11 PFAS could be detected in 85% of the samples. The mean value for the sum of PFAS for all samples was 2.96 µg/kg, the median was 1.25 µg/kg. The highest burden was found in one specimen of an ahanta francolin, at 27.84 µg/kg. The highest quantified concentration for one PFAS was 22 µg/kg of PFHxA, which also caused the high total load in this one specimen of ahanta francolin. PFOS was the Perfluorosulfonic acid (PFSA) with the highest detection rate in all samples (Table 2), as it could be detected in 54% and quantified in 44% of all samples. The only other PFSA that could be detected was PFHxS, which was found in two samples from pigs at levels below the limit of quantification. Neither PFBA nor PFPeA could be detected in any of the samples, as the combination of the two measurements did not produce any signals that met the identification criteria. Results for the individual samples are given in Table S5.

The analyte with the overall highest detection rate was PFHxA (75%). PFHxA is a short-chain perfluorocarboxylic acid (PFCA) and is primarily a byproduct of fluorotelomerization and a metabolite of other non-legacy PFAS produced by this process (Prevedouros et al. 2006; Backe et al. 2013). Due to its higher solubility, it exhibits greater mobility in the environment than long-chain PFCA; however, because of its rapid elimination, it has no known potential for bioaccumulation in the bodies of humans or animal (Russell et al. 2013; Numata et al. 2014; Guruge et al. 2016). Although the highest concentration within our study was detected in a sample from an omnivorous ahanta francolin (22 µg/kg), there is no significant difference between omnivorous and herbivorous species ($p = 0.7603$) or wild living and domesticated animals ($p = 0.5978$).

Short-chain PFCA show a better uptake by plants into their green parts like leaves, whereas long-chain PFCA tend to accumulate in the roots (Felizeter et al. 2012; Krippner et al. 2014). This may explain why PFHxA, unlike PFOA, PFOS, and PFAS with longer carbon chains, can be detected in the livers of duikers and sheep, which feed primarily on green plant parts (Jarman 1974; Parsons et al. 1994). PFHxA may also be more widespread in the environment around Kumasi than other PFAS because it occurs there as a metabolite of other non-legacy PFAS and has, as a result, contaminated the feed of the animals. However, this remains a hypothesis, as no information is available on PFAS exposure in the feed of the animals studied here, or on the environmental PFAS exposure in the Kumasi area.

The sum of PFAS found in the livers of domesticated animals was higher, but not statistically significantly higher,

Table 2 Overview of the PFAS levels found and their detection frequencies in the liver tissue of livestock (n = 14) and game animals (n = 27) from Kumasi, Ghana

Analyte	> LOD [%]	Mean [µg/kg]	Median [µg/kg]	Max. [µg/kg]	Min. [µg/kg]
PFBA	0	< LOD	< LOD	< LOD	< LOD
PFPeA	0	< LOD	< LOD	< LOD	< LOD
PFHxA	75	1.3	< LOQ	22	< LOD
PFHpA	40	0.27	< LOD	5.3	< LOD
PFOA	20	< LOD	< LOD	0.24	< LOD
PFNA	20	< LOQ	< LOD	0.66	< LOD
PFDA	25	< LOQ	< LOD	0.59	< LOD
PFUnDA	28	< LOQ	< LOD	0.44	< LOD
PFDoDA	15	< LOD	< LOD	0.27	< LOD
PFTrDA	20	< LOD	< LOD	0.2	< LOD
PFTeDA	15	< LOD	< LOD	0.24	< LOD
PFBS	0	< LOD	< LOD	< LOD	< LOD
PFPeS	0	< LOD	< LOD	< LOD	< LOD
PFHxS	5	< LOD	< LOD	< LOQ	< LOD
PFHpS	0	< LOD	< LOD	< LOD	< LOD
PFOS	55	0.70	< LOQ	370	< LOD
PFNS	0	< LOD	< LOD	< LOD	< LOD
PFDS	0	< LOD	< LOD	< LOD	< LOD
PFUnDS	0	< LOD	< LOD	< LOD	< LOD
PFDoDS	0	< LOD	< LOD	< LOD	< LOD
PFTrDS	0	< LOD	< LOD	< LOD	< LOD

than those found in the livers of wild animals. The mean value for domesticated animals was 4.15 µg/kg, while the mean value for wild animals was 2.33 µg/kg, but as mentioned the difference between the two groups was not significant ($p = 0.265$). This can be attributed to the impact of a small number of outliers with elevated levels of individual PFAS on the mean value of the group of domesticated animals. Moreover, it is not solely the proximity to humans that plays a role in the uptake of PFAS into the liver. As PFAS are adsorbed by the animal through their feed, omnivorous animals have the potential to accumulate large amounts of PFAS through sources such as carrion or prey (Ecke et al. 2025). In addition, it is likely that species-specific differences resulting from changes in protein expression play a role in the uptake and distribution of PFAS in tissues (Bangma et al. 2022). For the samples examined here, there is a higher, but not statistically significantly higher ($p = 0{,}130$), mean concentration of the sum of PFAS in omnivorous animals (3.84 µg/kg), compared to herbivorous animals (2.26 µg/kg).

The highest numbers of PFAS were detected in the livers of pigs and civets with eight to ten substances at concentrations above the limit of detection. Both species are distinguished by their omnivorous dietary habits (Angelici 2000). A significantly higher number of PFAS ($p < 0.01$) was detected in samples from omnivorous species (median = 4.5) than in samples from herbivorous species (median = 2).

Differences Between Studied Species

When interpreting the data, it is crucial to remember that different species with different habitats, husbandry and diets were analysed here. As Fig. 2 shows, the relative compositions of PFAS contamination vary between the different species. Grasscutters and Pigs showed the highest proportion of PFOS, while it was not detectable in Duikers. The highest numbers of different PFAS could be detected in Pigs and Civets, both omnivorous species with an opportunistic feeding behaviour (Angelici 2000; Cuevas et al. 2010). In contrast to herbivorous species, all substances from the group of long-chain PFCA (PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTrDA, PFTeDA) are detectable in these two species. When comparing the two groups of omnivorous and herbivorous animals, the concentrations of all long-chain PFCA in the samples from omnivorous animals are significantly higher than in the livers of herbivorous animals. No significant differences were found between wild and domesticated animals. The p-values for all substances can be found in Table S6. These findings are consistent with results that have been reported before, showing that omnivorous animals accumulate higher amounts of long-chain PFCA within their livers (Guckert et al. 2023).

Pigs are omnivorous farm animals that are kept in close proximity to humans. The samples of pigs that were analysed here originate from caged animals and were purchased in the area of Abrepo Junction. The mean total PFAS

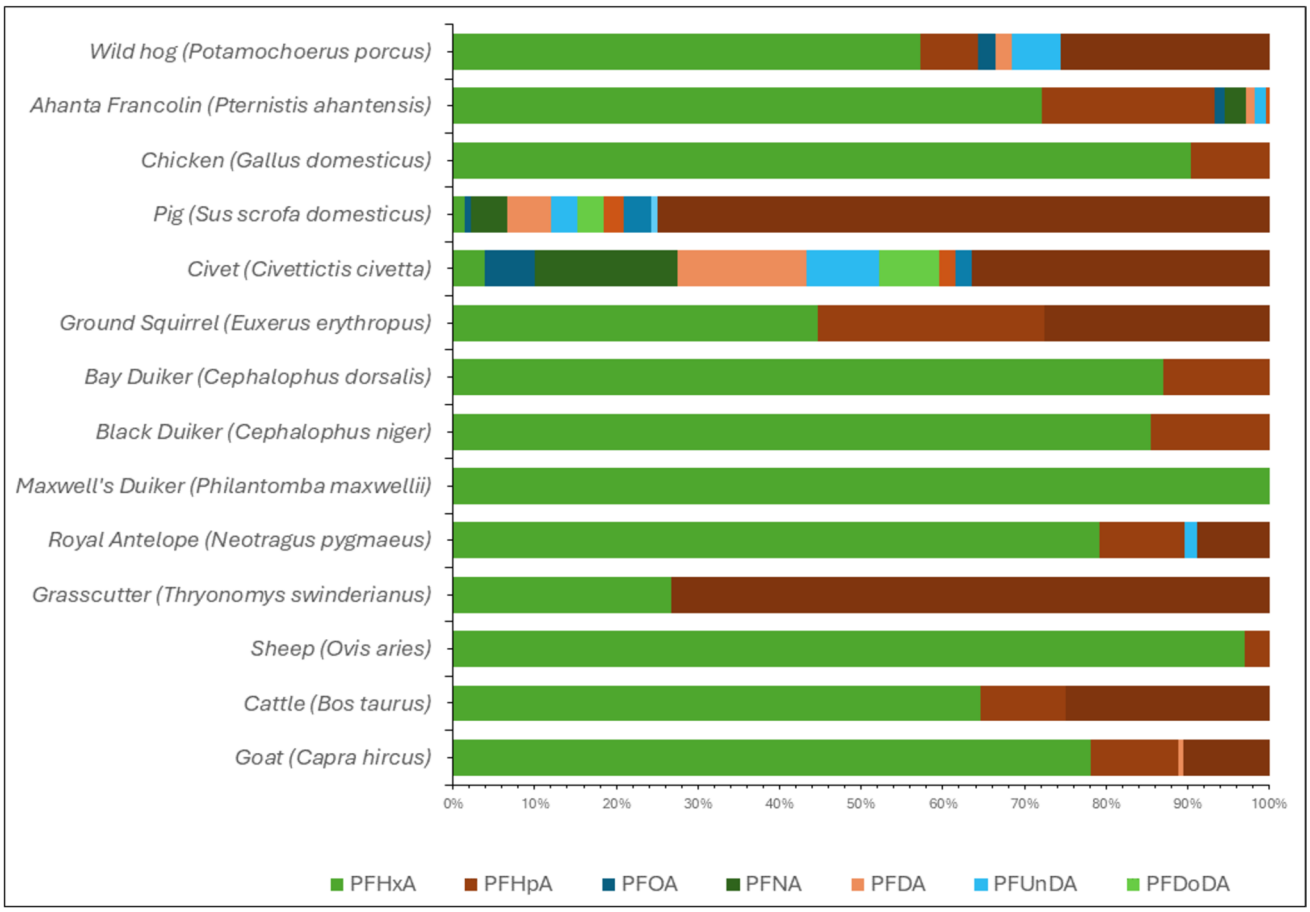


Fig. 2 Relative composition of the PFAS contamination in the livers of livestock and game animals from Kumasi, Ghana

contamination of the animals was 6.8±0.52 µg/kg (see Table S5). Even if the origin of the animals cannot be traced back from point of purchase to the producing farms, this small variation indicates a common origin of the animals with the same diet, as this is likely the main source of PFAS contamination of the animals. Duikers are a contrast to these domesticated animals. They lead a reserved life and tend to hide in forests and dense undergrowth. Their shy behavior means that little is known about the behaviour of these animals, but their diet consists largely of leaves and other plant parts (Jarman 1974). Their low levels of PFAS contamination are in line with what would be expected from their diet and general distance from humans and urbanised areas.

Comparison of the PFAS Contamination with Findings from Europe

When we compare the results of our study with those of other studies, it is noticeable that the level of contamination in the liver samples examined here is lower than that in the livers of animals from the European continent. It is not possible to compare these results with those from other samples of mammals from Ghana, as no such data has been published yet. Most published results focus on mammals from Europe or the Arctic Circle, as this area is often used as an example of a remote area with little anthropogenic influence. Table 3 compares results from studies conducted in European countries to results from the animals included in our investigation.

Many results have been reported for wild boars (*Sus scrofa)*, which often exhibit high levels of PFAS with liver concentrations above 100 µg/kg, and PFOS as the dominant substance (Felder et al. 2023; Moretti et al. 2023; Rupp et al. 2023; Schröder et al. 2024). Therefore, wild boars have also been proposed as bio indicators for PFAS in the environment (Kowalczyk et al. 2018). The findings are often linked to the omnivorous diet and digging habits of these animals. wild hogs, which, like wild boars belong to the family of *Suidae*, also exhibit this behavior pattern and furthermore their diet is comparable (Leslie and Huffman 2015). In the samples of wild hogs examined in this study, we were able to detect substantially lower levels of PFAS than it is known from studies of wild boars from Europe, as the mean PFAS content in our samples was only 2.1 µg/kg, while studies

Table 3 Comparison of median results for the occurrence of PFAS in liver tissue from animals in different countries. All results are expressed in µg/kg

Reference	Species	Country	PFHxA	PFHpA	PFOA	PFNA	PFDA	PFUnDA	PFDoDA	PFTrDA	PFTeDA	PFOS
(Tavoloni et al. 2023)	Wild boar (*Sus scrofa, n=26*)	IT	0.06	0.44	1.75	4.63	4.245	3.535	2.245	4.34	2.25	46.46
(Mertens et al. 2025)	Wild boar (Sus scrofa, n=82)	DE	0.12	0.4	1.78	4.36	3.17	2.04	1.93	–	2.05	81.4
(Schröder et al. 2024)	Wild boar (*Sus scrofa, n=29*)	PL	<2.5	1.2	22	68	19	4.1	2.8	8.4	2.6	51
(Draghi et al. 2025)	Pig (*Sus srofa domesticus, n=38*)	IT	1.302	3.488	37.301	6.910	2.305	0.534	0.240	0.135	0.068	<0.03
(Herzke et al. 2023)	Arctic Fox (*Vulpes lagopus, n=5*)	NOR	–	–	0.83	5.4	4.32	4.24	0.59	1.8	0.30	69.4
(Herzke et al. 2023)	Mink (*Neogale vison, n=5*)	NOR	–	–	1.79	27.4	15.9	18.5	2.45	2.89	0.37	264
(Herzke et al. 2023)	Moose (*Alces alces, n=5*)	NOR	–	–	<0.05	0.18	0.09	0.11	<0.05	<0.1	<0.1	0.27
(Herzke et al. 2023)	Wolf (*Canis lupus, n=5*)	NOR	–	–	<0.05	1.18	0.7	0.7	0.14	0.1	<0.1	1.81
(Müller et al. 2026)	Roe Deer (*Capreolus capreolus, n=28*)	AT	–	1.42	0.47	1.09	0.32	0.31	0.21	0.11	0.04	2.62
This study	Civet (*Civettictis civetta, n=3*)	GHA	0.2	<0.07	0.2	0.62	0.54	0.31	0.26	0.07	0.07	13
This study	Pig (*Sus scrofa domesticus, n=3*)	GHA	0.07	<0.07	0.07	0.31	0.35	0.2	0.2	0.2	0.2	4.7
This study	Duiker (*Cephalophini n=9*)	GHA	0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07
This study	Chicken *(Gallus domesticus, n=3)*	GHA	0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07
This study	*Ground Squirrel (Euxerus erythropus, n=3)*	GHA	0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	0.07
This study	*Wild hog (Potamochoerus porcus, n=3)*	GHA	0.42	<0.07	<0.07	<0.07	<0.07	0.07	<0.07	<0.07	<0.07	0.23
This study	*Ahanta francolin (Pternistis ahantensis, n=3)*	GHA	1.3	1.4	0.2	0.2	0.07	<0.07	<0.07	0.07	<0.07	<0.07
This study	*Grasscuttel (Thryonomys swinderianus, n=3)*	GHA	0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	0.67
This study	*Cattle (Bos taurus, n=3)*	GHA	1	0.31	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	1.1
This study	Goat *(Capra hircus, n=3)*	GHA	7.935	1.09	<0.07	<0.07	0.07	<0.07	<0.07	<0.07	<0.07	1.065
This study	*Sheep (Ovis aries, n=3)*	GHA	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07
This study	*Royal Antelope (Neotragus pygmaeus, n=3)*	GHA	1	0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	<0.07	0.07

Table 4 Estimated mean PFAS burden in the muscle meat of different species from Kumasi, Ghana

species	PFOA [µg/kg]	PFNA [µg/kg]	PFHxS [µg/kg]	PFOS [µg/kg]	sum [µg/kg]
Goat *(capra hircus, n=.2)*	–	–	–	0.24	0.24
Cattle *(bos taurus, n=3)*	–	–	–	0.26	0.26
Sheep *(ovis aries, n=3)*	–	–	–	–	–
Grasscutter *(thryonomys swinderianus, n=3)*	–	–	–	0.26	0.26
Antelope *(neotragus pygmaeus, n=3)*	–	–	–	0.03	0.03
Maxwell Duiker *(philantomba maxwellii, n=3)*	–	–	–	–	–
Black Duiker *(cephalophus niger, n=3)*	–	–	–	–	–
Bay Duiker *(cephalophus dorsalis, n=3)*	–	–	–	–	–
Ground Squirrel *(euxerus erythropus, n=3)*	–	–	–	0.02	0.02
Civet *(civettictis civetta, n=3)*	0.08	0.25	–	0.28	0.60
Pig *(sus scrofa domesticus, n=3)*	0.02	0.12	0.01	1.1	1.20
Chicken *(gallus domesticus, n=3))*	–	–	–	–	–
Ahanta Francolin *(pternistis ahantensis, n=3)*	0.05	0.11	–	–	0.16
Wild hog *(potamochoerus porcus, n=3)*	0.01	–	–	0.07	0.07

from Europe often show mean PFAS burdens often above 50 µg/kg (see Table 3). PFOS usually accounts for the largest proportion of total PFAS contamination in the livers of wild boars and, in studies conducted in Europe, often contributes around 80% of the total PFAS content. The proportion of PFOS in the samples from wild hogs here was only 14% of the total PFAS load. Mateus-Vargas et al. (2025) highlighted the influence of local land-use characteristics on PFAS-contamination in wild boar liver, showing that the county of origin significantly influences the PFAS concentration in the livers of wild boars (Mateus-Vargas et al. 2025). Felder et al. (2026) were able to show a correlation between the concentration of certain PFAS in the livers of wild boars and the proximity of potential point sources, such as landfills or wastewater treatment plants, even when all samples originated from one single German state (Felder et al. 2026). In the context of larger distances between habitats of wild boars in Germany, Rupp et al. (2023) highlighted the differences between animals living in close proximity of known point sources and those showing only the suspected background contamination (Rupp et al. 2023). In conclusion, this could suggest that global differences in environmental PFAS contamination are also reflected in the livers of species from the Suidae family.

For pigs, Draghi et al. (2025) published data on 38 pigs from Northern Italy. In their study, PFAS could be found in 100% of the investigated samples with a mean concentration of the sum of PFCA above 200 µg/kg (dominated by PFBA) and a mean value for PFHxS of 311 µg/kg, which was the most abundant PFSA as they reported no concentrations of PFOS above the limit of quantification in their study (Draghi et al. 2025). PFBA could not be detected in the samples analysed here, as we were unable to identify any peaks that met the identification criteria in both measurements performed. For long chain PFCA, the concentrations found in their samples were between 2 times (PFTeDA) and 1676 times (PFOA) higher than our findings. These results support the hypothesis that environmental PFAS contamination levels in Ghana are lower than in Europe, leading to lower uptake and accumulation in the livers of the pigs.

Relevance of the Findings for Consumer Safety in Ghana

As no maximum levels for PFAS in food have yet been set in Ghana, classification can only be based on other assessments. The European basis for assessing the PFAS content of food is the current EFSA assessment of PFOA, PFNA, PFOS, and PFHxS and the resulting maximum levels in Annex I of the Commission Regulation (EU) 2023/915 on maximum levels for certain contaminants in food and repealing Regulation (EC) No 1881/2006. The Annex sets out maximum levels for meat and offal (Commission Regulation (EU) 2023/915). The concentrations in all samples were below the maximum levels. For domesticated animals, the sum of the four regulated PFAS corresponded on average to 19% of the maximum levels, while for game it is 0.93%. For pigs the sum of the four regulated PFAS equals 64% of the maximum level, which is the highest value for one species from our study.

Cartron et al. have proposed a correlation between the PFAS burden in the liver and the muscle meat of mammals and birds by determining the correlation between the concentrations in the liver and muscle tissue across different species of birds and mammals for the four substances PFOA, PFNA, PFHxS, and PFOS. (Cartron et al. 2025). Using the conversion factors given in Table S7 to apply these findings to our results leads to the estimated mean muscle concentrations presented in Table 4. In this assessment, only the calculated PFOS level in pigs (1.1 µg/kg)

exceeds the maximum level of 0.3 µg/kg set for meat of bovine animals, pigs and poultry by the Commission Regulation (EU) 2023/915.

Although the calculated value exceeds the maximum level set by these regulations, it should be noted that such a calculation is only an indication and cannot replace a direct determination of PFAS contamination in the muscle meat. Furthermore, the limits established for Europe are based on European consumption habits, and per capita meat consumption in Ghana is lower than in Europe (FAOSTAT 2025). However, the calculations indicate that it is also reasonable to consider PFAS exposure through animal-based foods for the people of Ghana, and that further research is necessary.

Conclusion

In our study, we were able to present data on PFAS contamination in mammals in Ghana for the first time. Across all substances examined, concentrations were lower than those previously reported for comparable samples from the European continent. When comparing the species studied, omnivorous animals show a higher diversity of quantifiable PFAS, which is consistent with a potential accumulation within the food chain. The impact of PFAS on food safety cannot be ignored, but can only be estimated due to the small number of samples and the derivation of possible contamination in muscle meat. Although the number of samples studied is small, this study highlights the importance of conducting investigations on the African continent when considering global PFAS exposure. In this context liver samples, especially from omnivorous species, can be a useful tool for comparing different habitats.

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Data Availability The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format they are available from the corresponding author upon reasonable request.

Declarations

Competing Interests The authors have not disclosed any competing interests.

Ethical Approval Liver samples were obtained from animals legally hunted or slaughtered for food production. No animal was killed for the purposed of being included in this study, therefore no ethical committee approval was required.

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